

STAT 230: Probability

Lecture 4

Today's Agenda

Last Lecture:

- ▶ Odds of an event
- ▶ Equally likely sample spaces
- ▶ Notation
- ▶ Permutations vs combinations

Today:

- ▶ Permutations vs combinations
- ▶ Multinomial coefficient

Motivation - Permutations

Suppose that the 5 members of the UW hardcour parkour club must select a president and vice-president. How many ways can they do this?

$$P(5, 2) = \frac{5!}{2!(5-2)!} = \frac{5!}{2!3!} =$$

Counting Permutations or Ordered Arrangements

Given n distinct objects, a **permutation** of size k is an *ordered* subset of k of the objects. The number of permutations of size k taken from n objects is denoted $n^{(k)}$ and

$$n^{(k)} = n(n-1)\dots(n-k+1) = \frac{n!}{(n-k)!}.$$

- ▶ $n^{(k)} =$ “ n to k factors”, $k > 0$
- ▶ $n^{(0)} = 1$
- ▶ $n^{(n)} = n!$ and $0^{(0)} = 0! = 1$
- ▶ The tricky part of this definition is the word “ordered”. An ordering need not be numerical, for example assigning labels like “President” and “Vice-President” has the effect of ordering the individuals.

Question - Permutations

Which of the following situations would be considered a permutation?

- A Draw a hand of 5 cards from a deck
- B Select a committee of 3 people from a class
- C 3 people getting matched with 3 co-op employers
- D Choose 4 products to test for defects
- E None of the above

Example - Selecting Numbers

Suppose we randomly select three numbers **without** replacement from $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ to form a three digit number. What is the probability that:

1. The number is larger than 500?

$$\underline{\quad} \underline{\quad} \underline{\quad} \quad \text{are } \{5, 6, \dots, 9\} = 5$$
$$\begin{array}{c} 5^{(1)} \quad . \quad 8^{(2)} \\ \hline 9^{(3)} \end{array}$$

2. The number is even?

$$\begin{array}{c} 4^{(1)} \quad . \quad 8^{(2)} \\ \hline 9^{(3)} \end{array}$$

Example - Selecting Numbers (continued)

Suppose we randomly select three numbers **without** replacement from $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ to form a three digit number. What is the probability that:

3. The number is larger than 700, and is even?

▶
$$\begin{array}{r} 2^1 \quad 7^1 \cdot \quad 4^1 \quad + \quad 7^1 \cdot 3^1 \\ \hline 9(3) \end{array}$$

▶

▶

Motivation - Combinations or Number of Sets

Suppose instead that the 5 members of the UW hardcour parkour club must instead simply select two members to be on the budget committee. How many ways can they do this?

Counting Combinations or Number of Sets

Given n distinct objects, a **combination** of size k is an *unordered* subset of k of the objects. The number of combinations of size k taken from n objects is denoted by

$$\binom{n}{k} = \frac{n^{(k)}}{k!} = \frac{n!}{(n-k)!k!}.$$

- ▶ $\binom{n}{k}$ = “ n choose k ”
- ▶ $\binom{n}{k} = 0$ if $k < 0$ and $\binom{n}{0} = 1$
- ▶ This also known as the *Binomial Coefficient*
- ▶ For $n \in \mathcal{Z}^+$ and $k \leq n$

$$\binom{n}{k} = \frac{n^{(k)}}{k!} \times \frac{(n-k)!}{(n-k)!} = \frac{n!}{k!(n-k)!} = \binom{n}{n-k}$$

Question - Combinations

Which of the following is an incorrect application of combinations?

- A There are $\binom{49}{6}$ possible plays at the lotto 6/49.
- B There are $\binom{52}{5}$ poker hands at five-card draw poker.
- C There are $\binom{10}{6}$ ways that 6 heads can be obtained in a series of 10 coin flips.
- D There are $\binom{32}{10}$ ways that 10 distinguishable balls can be drawn without replacement from a bag of 32 balls.
- E There are $\binom{200}{190}$ ways that 190 distinct passengers can sit in a plane with 200 seats.

Example - Distinct Books

Suppose you have 20 distinct books and 7 were written by George R.R. Martin.

1. How many ways can you arrange 12 books on a shelf if the order they are on the shelf matters?

$$P(20, 12) = \frac{20!}{8!}$$

2. How many ways can you pick 12 books to put on a shelf (the order does not matter)?

$$\binom{20}{12} = \frac{20!}{8!12!}$$

3. How many ways can you select 12 books to put on a shelf if exactly 3 of them must be Martin books?

$$\binom{7}{3} \binom{13}{9}$$

4. A monkey picks books at random from the 20 books and puts them on the shelf until it contains 12 books. What is the probability that at least 3 of the books on the shelf are written by Martin?

$$\begin{array}{r}
 \binom{7}{0} \cdot \binom{13}{12} + \binom{7}{1} \cdot \binom{13}{11} + \binom{7}{2} \cdot \binom{13}{10} \\
 \hline
 20 \\
 12
 \end{array}$$

Motivation Rearranging - Letters or Indistinguishable Objects

- ▶ How many ways can we rearrange the letters "BIRD" to form a 'word'.

$$4!$$

- ▶ How many ways can we rearrange the letters "BRRR" to form a 'word'.

$$\frac{4!}{3!}$$

- ▶ How many ways can we rearrange the letters "DODO" to form a 'word'.

$$\frac{4!}{2! \cdot 2!}$$

- ▶ How many ways can we rearrange the letters "PEPPER" to form a 'word'.

$$\frac{6!}{3! \cdot 2!}$$

- ▶ How many ways can we rearrange the letters "ABBOZZO" to form a 'word'.

$$\frac{7!}{2! \cdot 2 \cdot 2}$$

Multinomial Coefficient

- ▶ Consider n objects which consist of $k \geq 2$ distinguishable types.
- ▶ Suppose that there are n_1 objects which are of type 1, n_2 which are of type 2, and in general n_i objects of type i so that $n_1 + n_2 + \cdots + n_k = n$.
- ▶ Then there are

$$\binom{n}{n_1 n_2 \cdots n_k} = \frac{n!}{n_1! n_2! \cdots n_k!}$$

arrangements of the n objects. This quantity is known as a **multinomial coefficient**.